Math Hallucination Detector: Detecting and Explaining Mathematical Hallucinations in Large Language Models

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Abstract

Large language models frequently produce answers that appear mathematically plausible but are incorrect, posing challenges for reliability in educational, scientific, financial, and other high stakes decision-making contexts. The **Math Hallucination Detector** is a system designed to automatically identify, verify, and explain errors in mathematical responses generated by LLMs.

The framework validates model outputs against authoritative ground-truth solutions obtained from Wolfram Alpha and applies deterministic numeric and symbolic comparison rules to detect inconsistencies. To improve reliability, the system incorporates reasoning based on first-order predicate calculus, enabling formal correctness checks, structured inference traces, and hypotheses about potential sources of error.

For equation based claims, the system performs step-by-step expression evaluation and backward error analysis to pinpoint divergence within sub expressions. This allows the detector to localize where incorrect reasoning or computation likely occurred.

The system produces clear verdicts, confidence scores, structured reasoning logs, and explanatory feedback for each evaluated response. By combining external computational verification with rule based logical reasoning, the **Math Hallucination Detector** provides a transparent and auditable framework for detecting and analyzing mathematical hallucinations in large language models.

Why It Matters

Educational reliability: Students and educators benefit from tools that not only flag incorrect solutions but also explain where the reasoning fails, supporting learning rather than simply marking answers wrong.

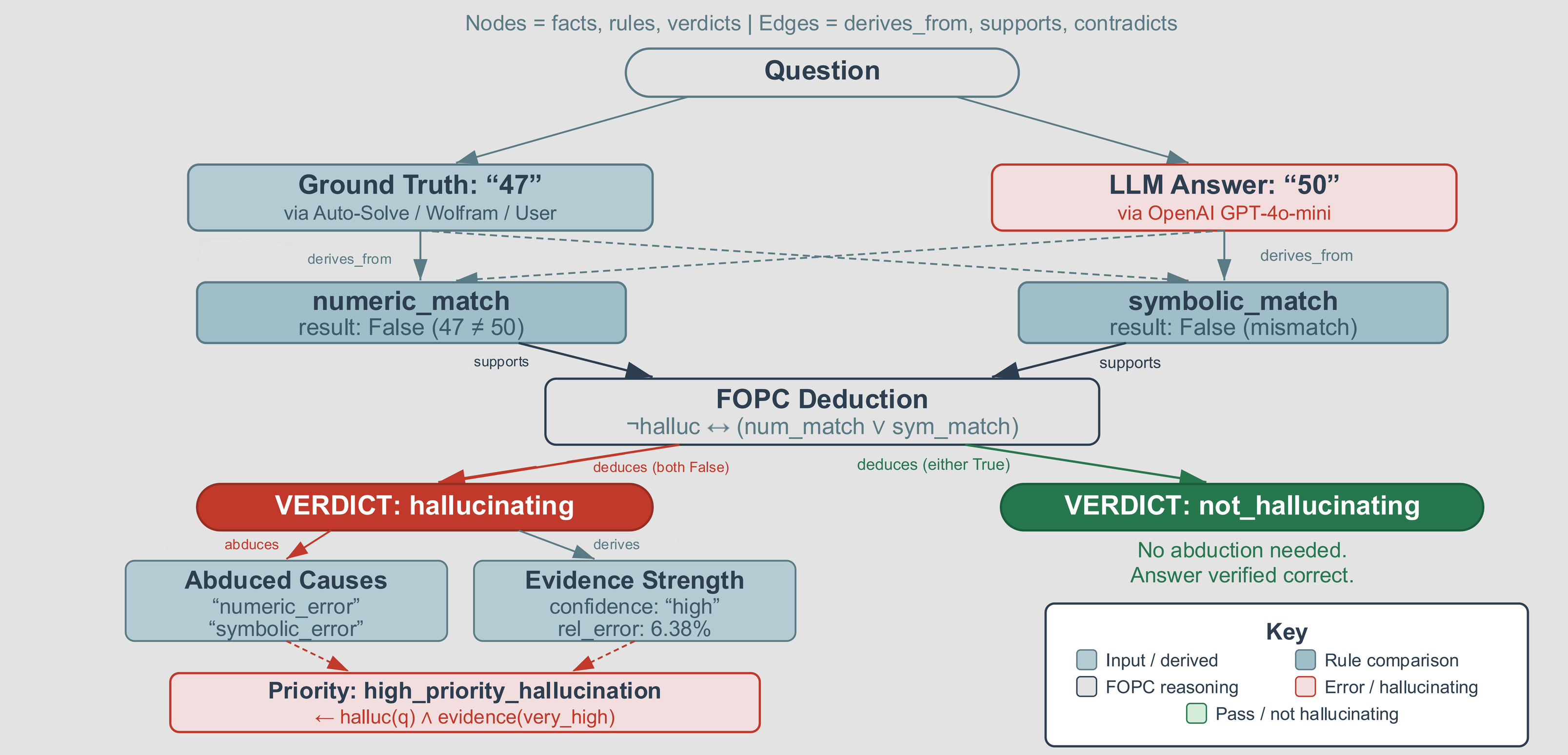
Trust in AI systems: Large language models are increasingly used for learning, research, and decision making. Detecting mathematical hallucinations helps ensure users can trust the correctness of AI-generated answers.

Transparency and accountability: Our system produces auditable reasoning traces and deterministic verdicts, making AI outputs easier to verify, debug, and systematically analyze for errors, inconsistencies, and points where reasoning diverges.

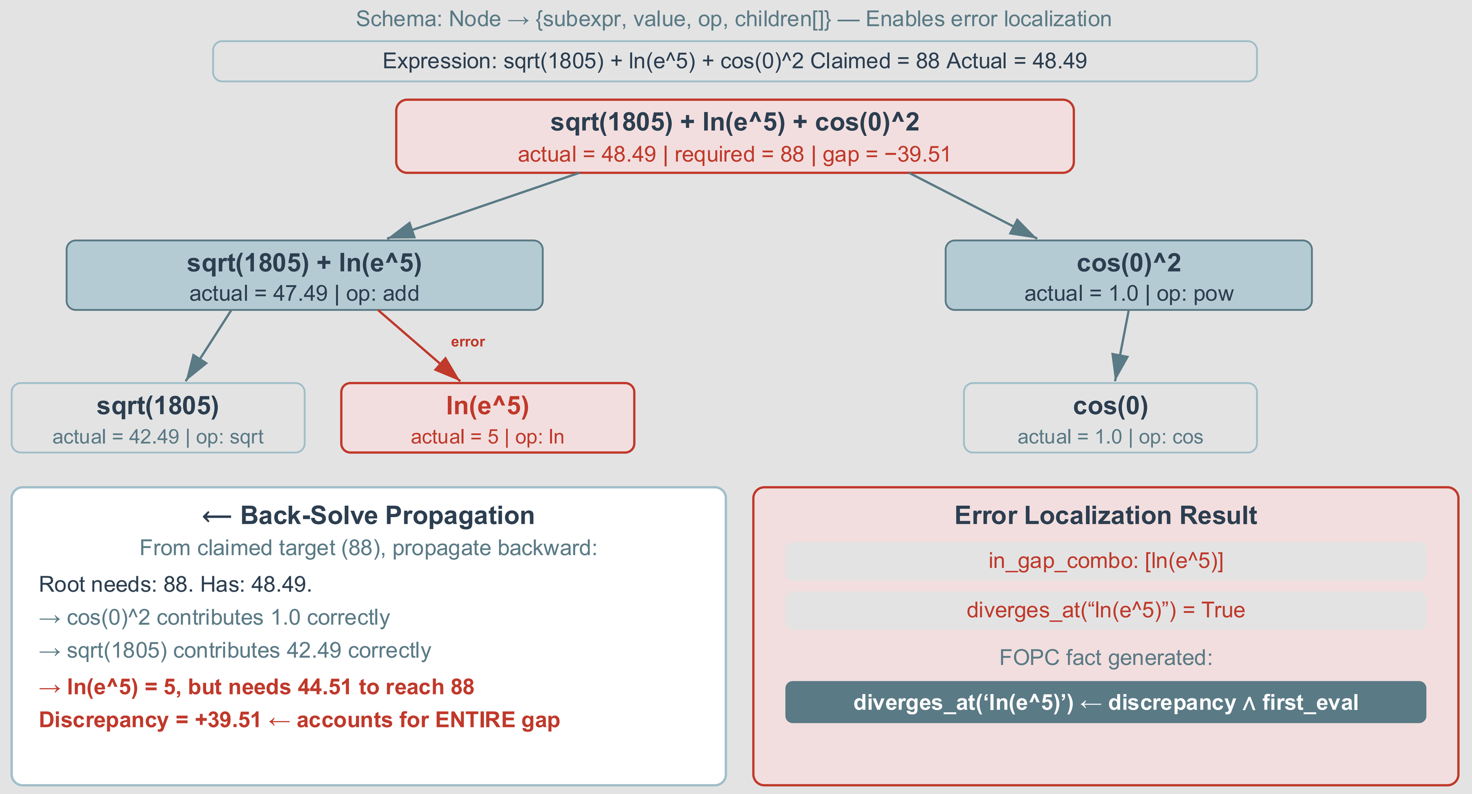
Scalable verification: By automatically generating and validating ground truth, the system reduces reliance on human reviewers and enables large-scale evaluation of LLM mathematical reasoning.

Model evaluation and improvement: By identifying patterns in errors and hallucinations, researchers and developers can better understand model limitations and use these insights to improve training methods, evaluation benchmarks, and future model architectures.

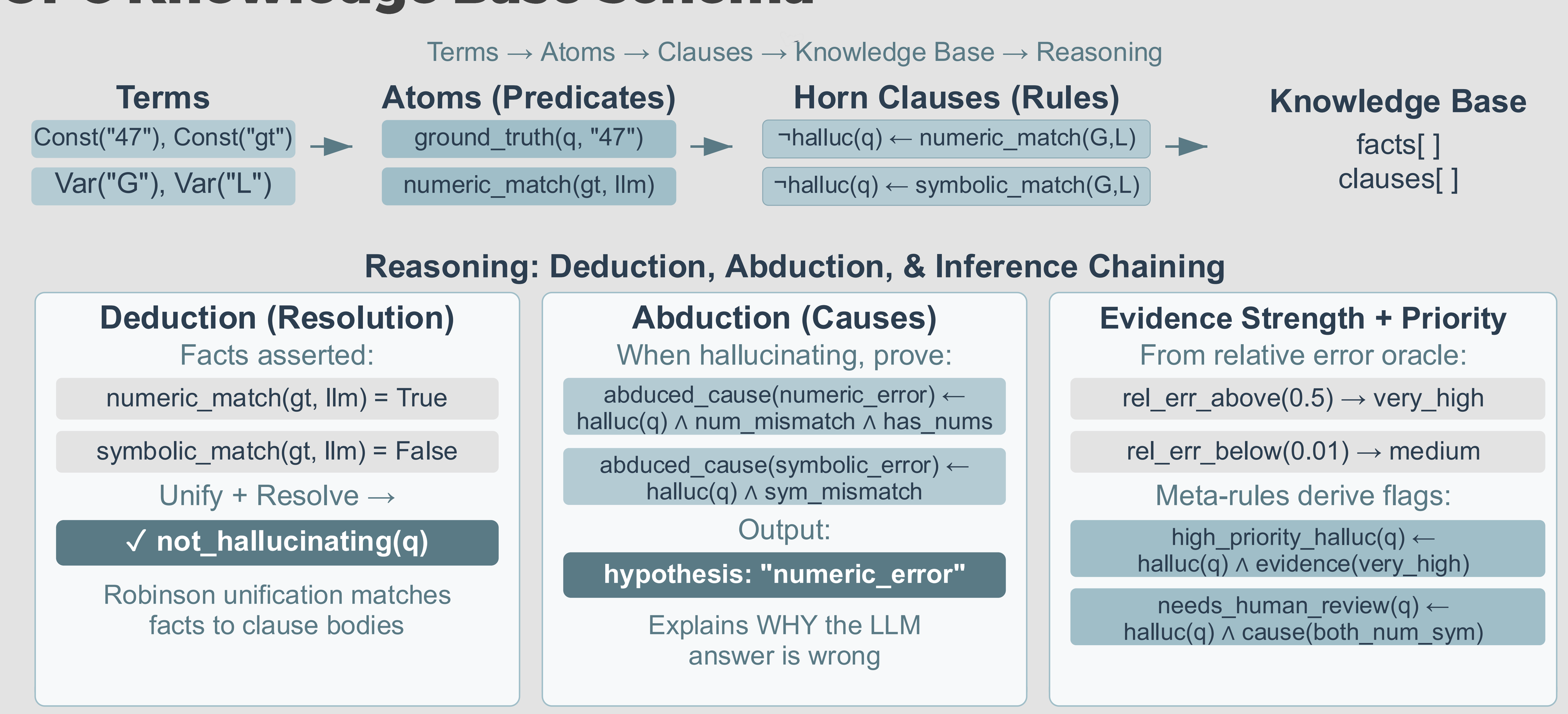
Reasoning Trace as Knowledge Graph



Reasoning Trace as Expression Tree



FOPC Knowledge Base Schema



Methods and How Reasoning Works

PIPELINE OVERVIEW

1 **Ground Truth** — Wolfram Alpha → local rules → LLM solver
→ 2 **LLM Answer** — Final answer + optional reasoning via API
→ 3 **Compare** — Apply numeric & symbolic comparison rules
→ 4 **Reason** — Deduce verdict; abduce causes; assign confidence
→ 5 **Output** — Verdict + FOPC trace + narrative explanation

Fixed sequence, no agentic loop. All reasoning is rule-driven and deterministic.

GROUND TRUTH SOURCES

1 PRIMARY **Wolfram Alpha**

Canonical answer, parsed to single value; step-by-step in equation mode

↓ *if fails or unit mismatch*

2 FALLBACK **Local pattern rules**

Hand-crafted for known failure modes (unit conversion, ratio questions, which quantity is asked)

↓ *if no pattern matches*

3 LAST RESORT **LLM solver**

Translates problem to evaluable expression; safely evaluated as ground truth only when needed

Local rules **override** Wolfram when they match — designed to fix known Wolfram failure modes. LLM-based ground truth is a last resort only.

LOGICAL REASONING (FOPC)

```
// Deduction - verdict rule
not_hallucinating ←
  numeric_match ∨ symbolic_match
// Meta-rule override
hallucinating ←
  numbers_clearly_differ ∧
  only_symbolic_match
------------------------------------
// abduction + confidence rules below
// priority meta-rules assign flags
```

[#]	[S]	[!]
Numeric error	**Symbolic error**	**Both**
Wrong number, op, or rounding	Wrong form or notation	Numeric + symbolic mismatch

When verdict = **hallucinating**, abduction infers the cause from which rule failed. Relative error sets **confidence level** (high / medium). Priority meta-rules flag cases needing human review.

WORKED EXAMPLE — WORD PROBLEM

"Three siblings share $90 equally. How much does each get?"

GROUND TRUTH $30 (Wolfram Alpha)

	CORRECT LLM		HALLUCINATING LLM
LLM ANS.	$30	LLM ANS.	$45
NUMERIC	match	NUMERIC	no match
VERDICT	NOT hallucinating	VERDICT	HALLU-CINATING

```
abduced_cause(numeric_error).
// used 90/2 instead of 90/3
confidence(high). // rel_err = 0.50
```

[CSV]
Tabular Output
One row per question: question, ground truth, LLM answer, numeric_match, symbolic_match, verdict → exportable as CSV / JSON for batch analysis.

[|-]
FOPC Trace
Facts + inferred predicates recorded so every conclusion is tied to premises and named rules. Fully auditable reasoning chain.

[TXT]
Narrative Explanation
Human-readable text built from the same trace — rule outcomes, verdict, abduced causes, and confidence. No free-text interpretation.

Conclusion

The **Math Hallucination Detector** shows that combining external computational verification with rule based logical reasoning offers a reliable way to detect mathematical hallucinations in LLM outputs. By validating answers against authoritative ground truth and applying deterministic numeric and symbolic comparison rules, the system can automatically identify discrepancies between generated and correct solutions.

Beyond simple answer checking, the framework produces structured reasoning traces, expression level evaluations, and FOPC based deductions that localize where reasoning diverges and infer likely causes of error. This enables transparent, auditable verification rather than opaque correctness judgments.

Overall, the project demonstrates that integrating symbolic reasoning, formal logic, and external computation can significantly improve reliability and interpretability of AI generated mathematical reasoning.

Additionally, the system provides a practical framework for systematically analyzing the reasoning behavior of large language models. By exposing intermediate evaluation steps and highlighting points of divergence, the detector enables more precise debugging of model outputs and offers insights into common patterns of mathematical hallucination. This capability can support both researchers and developers in improving model evaluation methodologies and designing more robust AI systems.

Future Work: Extending the framework to support broader mathematical domains and large-scale benchmark evaluation of LLM reasoning.